

# Is Presence of ANCA in Crescentic IgA Nephropathy a Coincidence or Novel Clinical Entity? A Case Series

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**Background:** There are few anecdotal reports of circulating antineutrophil cytoplasmic autoantibodies (ANCAs) in patients with immunoglobulin A (IgA) nephropathy.

**Study Design:** Retrospective case series.

**Setting & Participants:** We studied 8 patients with crescentic IgA nephropathy associated with ANCAs against myeloperoxidase ( $n = 5$ ) and proteinase 3 ( $n = 3$ ) followed up for  $2.4 \pm 1.7$  years. They were compared with 26 patients with IgA nephropathy with  $> 10\%$  crescentic glomeruli, but negative for ANCAs.

**Outcomes:** We analyzed clinical and histologic features of patients and their response to treatment.

**Measurements:** Screening for ANCAs was performed using indirect immunofluorescence, and positive results were verified using enzyme-linked immunosorbent assay.

**Results:** All patients with crescentic IgA nephropathy and positive for ANCAs, compared with only one-third of ANCA-negative patients, presented with the clinical syndrome of rapid progressive glomerulonephritis. ANCA-positive patients reached a higher peak serum creatinine level within the first 3 months ( $4.2 \pm 2.2$  vs  $2.5 \pm 1.9$  mg/dL; estimated glomerular filtration rate,  $19.3 \pm 10.2$  vs  $45.9 \pm 30.1$  mL/min/1.73 m<sup>2</sup>). ANCA-positive patients with IgA nephropathy had a higher percentage of crescentic glomeruli ( $54.3\% \pm 18\%$ ) compared with ANCA-negative patients with crescentic IgA nephropathy ( $34.5\% \pm 26\%$ ). ANCA-positive patients were treated using cyclophosphamide and corticosteroids. Kidney function improved in all these patients: serum creatinine level decreased from the peak of  $4.2 \pm 2.2$  to  $1.7 \pm 0.7$  mg/dL at the end of follow up (estimated glomerular filtration rate,  $19.3 \pm 10.2$  to  $44.6 \pm 11.1$  mL/min/1.73 m<sup>2</sup>). In contrast, no significant improvement was achieved in ANCA-negative patients.

**Conclusion:** Patients with IgA nephropathy, crescents, and positive for ANCAs represent a clinical entity with a diverse more exaggerated clinical and histologic picture. However, disease in these patients responded well to aggressive immunosuppressive therapy.

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**INDEX WORDS:** IgA nephropathy; antineutrophil cytoplasmic autoantibodies (ANCAs); rapidly progressive glomerulonephritis; crescents.

Immunoglobulin A (IgA) nephropathy, the most prevalent form of primary glomerulonephritis worldwide, is characterized by mesangial cell proliferation, matrix expansion, and mainly mesangial IgA deposition, often in conjunction with C3 and IgG and/or IgM. Despite its high frequency, the cause and pathogenesis of IgA nephropathy are enigmatic. Altered metabolism of polymeric IgA (production decreased in the mucosa<sup>1</sup> and increased in bone marrow<sup>2</sup>) and impaired IgA glycosylation and clearance<sup>3</sup> have been implicated. The clinical course of IgA nephropathy is very variable; although most patients have stable or slowly decreasing kidney function over many years, a subset of patients with a more progressive course has been described ("malignant IgA nephropathy"). In many cases, the histologic equivalent of the clinical picture of rapidly progressive IgA nephropathy is the presence of focal or diffuse necrotizing crescentic glomerulonephritis.<sup>4</sup> Recent prospective

studies showed that 40% of patients with as little as 10% cellular crescents progressed to end-stage renal disease within 3 years.<sup>5</sup>

There are few anecdotal reports of circulating antineutrophil cytoplasmic autoantibodies

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(ANCAs) in patients with crescentic IgA nephropathy.<sup>6-15</sup> In the present study, we report a cohort of patients with crescentic IgA nephropathy associated with ANCAs. We describe their clinical and histologic picture and response to treatment. Through comparison with patients with ANCA-negative crescentic IgA nephropathy, we try to address the question of whether the presence of ANCAs is merely a coincidence or rather defines a novel clinical entity.

## METHODS

Clinical and histologic data from 393 patients with IgA nephropathy were retrospectively analyzed. The diagnosis of IgA nephropathy was made using kidney biopsy between 1992 and 2007 at either the Department of Nephrology of the Heinrich-Heine University of Düsseldorf, Germany, or the Department of Nephrology of the Aristotelian University of Thessaloniki, Greece. All patients undergoing kidney biopsy were screened for ANCAs using indirect immunofluorescence. In the case of a positive result, verification using a standard enzyme-linked immunosorbent assay (ELISA) procedure was performed. Biopsy reports were reviewed retrospectively, and the presence of crescent formations was documented. The diagnosis of IgA nephropathy was based on the presence of dominant or codominant staining for IgA in the glomerular mesangium. Lupus nephritis was excluded based on clinical and serologic data.

A total of 8 patients were found to have ANCAs (against myeloperoxidase [MPO],  $n = 5$ ; against proteinase 3 [PR3],  $n = 3$ ). All 8 patients had crescents in  $> 10\%$  of glomeruli in the biopsy specimen. They were followed up for  $2.4 \pm 1.7$  years. Monitoring included serum creatinine level, proteinuria, blood pressure, and immunosuppressive and antihypertensive medication. Estimated glomerular filtration rate (eGFR) was calculated for each patient visit using the 4-variable Modification of Diet in Renal Disease Study equation.

In our cohort of 393 patients with biopsy-proven IgA nephropathy, 26 patients also were found to have  $> 10\%$  crescentic glomeruli, but were negative for ANCAs. These patients were analyzed as a control group, and their clinical and histologic findings were compared with those of patients with ANCA-positive IgA nephropathy.

Kidney biopsy specimens from both ANCA-positive and ANCA-negative patients were analyzed and the following features were recorded (among others). In light microscopy, the proportion of obsolescent glomeruli, proportion of glomeruli with cellular or fibrous crescents, degree of mesangial matrix expansion and mesangial cell proliferation (estimated semiquantitatively on a scale of 0-3 [0 for absence, + for mild, ++ for moderate, and +++ for severe]), and degree of tubular atrophy (expressed as percentage of tubules per glomerular cross-section that showed the characteristic findings of atrophy). In immunohistochemistry, the presence of IgA, IgG, IgM, C3, C4, C1q, and fibrin deposits was documented (also graded on a scale of 0-3). Localization of deposits was verified using electron microscopy.

## RESULTS

### Baseline Clinical Characteristics

#### *ANCA-Positive Patients With IgA Nephropathy*

In the present study, we analyzed 8 patients with biopsy-proven IgA nephropathy who also had circulating ANCAs. Five patients were positive for MPO-ANCA, and 3 patients, for PR3-ANCA. All other test results for autoantibodies (especially antinuclear, anti-DNA, and anti-glomerular basement membrane antibodies) were negative, and complement levels were within the normal range. Two patients were women and 6 were men. Mean age at diagnosis was  $56.0 \pm 6.3$  years. All 8 patients presented with rapidly progressive glomerulonephritis and fast deteriorating kidney function. Three patients presented with macroscopic hematuria, and all except 1 had systemic symptoms, such as weakness, arthralgias, or fever (Table 1). Six patients had hemoptysis and/or pulmonary infiltrates (Table 1). No patient had cutaneous purpuric lesions. As shown in Fig 1 (dark line), mean serum creatinine level at presentation was  $2.7 \pm 2.6$  mg/dL (MPO-ANCA,  $3.6 \pm 2.9$  mg/dL; PR3-ANCA,  $1.2 \pm 0.2$  mg/dL). As shown in Fig 2 (dark line), mean eGFR was  $49.3 \pm 31.7$  mL/min/1.73 m<sup>2</sup> (MPO-ANCA,  $35.9 \pm 31.9$  mL/min/1.73 m<sup>2</sup>; PR3-ANCA,  $71.7 \pm 16.5$  mL/min/1.73 m<sup>2</sup>). Within days or weeks, serum creatinine level increased to a maximum of  $4.2 \pm 2.2$  mg/dL (MPO-ANCA,  $4.8 \pm 2.5$  mg/dL; PR3-ANCA,  $3.1 \pm 1.0$  mg/dL) and eGFR decreased to a minimum of  $19.3 \pm 10.2$  mL/min/1.73 m<sup>2</sup> (MPO-ANCA,  $16.2 \pm 10.7$  mL/min/1.73 m<sup>2</sup>; PR3-ANCA,  $24.4 \pm 8.6$  mL/min/1.73 m<sup>2</sup>). All patients had active urinary sediment with hematuria and mean proteinuria with protein excretion of  $3.9 \pm 2.4$  g/24 h (range, 1.4-7.3 g/24 h). Three patients had clinical overt nephrotic syndrome. Seven of 8 patients were hypertensive and required antihypertensive treatment (mean arterial blood pressure,  $105.5 \pm 9.1$  mm Hg). Mean hemoglobin level at presentation was  $11.1 \pm 1.9$  g/dL and decreased to a minimum of  $10.5 \pm 2.0$  g/dL during the in-hospital evaluation.

#### *Comparison With Patients With ANCA-Negative Crescentic IgA Nephropathy*

Table 2 shows the comparison of clinical data between ANCA-positive ( $n = 8$ ) and ANCA-

Table 1. Clinical Characteristics of ANCA-Positive Patients With Immunoglobulin A Nephropathy

| Patient No. | ANCA | Sex | Age (y) | Fever | Respiratory Involvement           | Arthralgias | Macrohematuria | Initial SCr (mg/dL) | Initial eGFR (mL/min/1.73 m <sup>2</sup> ) | Peak SCr (mg/dL) | Lowest eGFR (mL/min/1.73 m <sup>2</sup> ) | Proteinuria (mg/24 h) |
|-------------|------|-----|---------|-------|-----------------------------------|-------------|----------------|---------------------|--------------------------------------------|------------------|-------------------------------------------|-----------------------|
| 1           | MPO  | F   | 58      | +     | Pulmonary infiltrates             | —           | +              | 1.5                 | 37.9                                       | 3.6              | 13.8                                      | 6,600                 |
| 2           | MPO  | M   | 59      | +     | Sinusitis                         | —           | +              | 1.9                 | 40.0                                       | 3.3              | 20.5                                      | 7,300                 |
| 3           | MPO  | M   | 62      | —     | —                                 | —           | +              | 6.4                 | 9.4                                        | 6.7              | 9.0                                       | 1,600                 |
| 4           | MPO  | F   | 60      | +     | —                                 | —           | —              | 7.1                 | 6.3                                        | 8.2              | 5.3                                       | 3,600                 |
| 5           | MPO  | M   | 61      | +     | Hemoptysis                        | —           | —              | 1.0                 | 85.7                                       | 2.2              | 32.5                                      | 3,360                 |
| 6           | PR3  | M   | 53      | +     | Pulmonary infiltrates, hemoptysis | —           | —              | 1.1                 | 74.4                                       | 2.2              | 33.4                                      | 5,700                 |
| 7           | PR3  | M   | 53      | +     | Pulmonary infiltrates             | +           | —              | 1.5                 | 54.1                                       | 4.1              | 16.3                                      | 1,400                 |
| 8           | PR3  | M   | 43      | +     | Hemoptysis                        | +           | —              | 1.0                 | 86.7                                       | 3.1              | 23.5                                      | 1,600                 |

Note: Conversion factors for units: SCr in mg/dL to  $\mu\text{mol/L}$ ,  $\times 88.4$ ; eGFR in mL/min/1.73 m<sup>2</sup> to mL/s/1.73 m<sup>2</sup>,  $\times 0.01667$ .

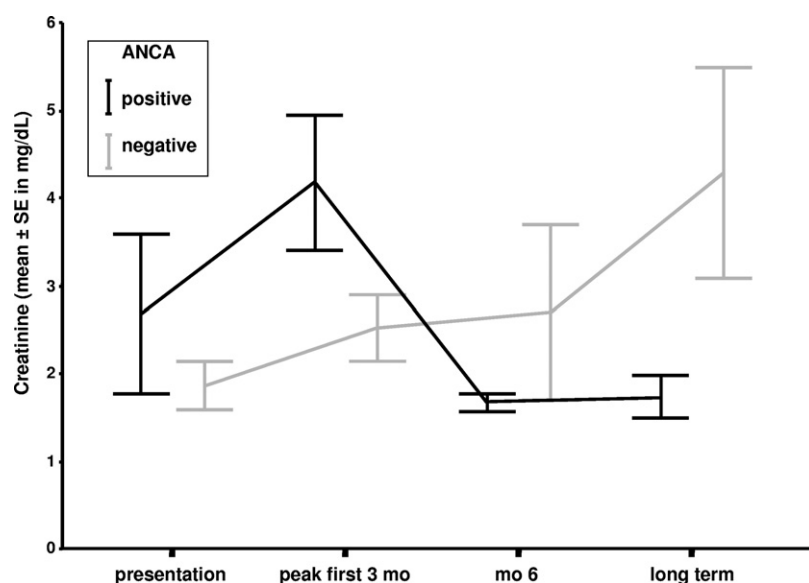
Abbreviations: ANCA, antineutrophil cytoplasmic antibody; eGFR, estimated glomerular filtration rate; MPO, myeloperoxidase; PR3, proteinase 3; SCr, serum creatinine.

negative ( $n = 26$ ) patients with biopsy-proven IgA nephropathy and  $> 10\%$  crescentic glomeruli. ANCA-positive patients were significantly older and more often had systemic signs compared with ANCA-negative patients. They also tended to have worse kidney function at presentation. In the first 3 months after diagnosis, ANCA-positive patients experienced more severe deterioration in kidney function (Table 2; Figs 1 and 2). Hemoglobin levels were lower in ANCA-positive patients. Furthermore, ANCA-positive patients had higher proteinuria levels and more often had macroscopic hematuria (Table 2). Overall, ANCA-positive patients presented with a more severe clinical picture: all of them (compared with only one-third of ANCA-negative patients) presented with the clinical syndrome of rapidly progressive glomerulonephritis (Table 2).

## Histologic Features

### ANCA-Positive Patients With IgA Nephropathy

Mean percentage of glomeruli with crescent formation was  $54.3\% \pm 18\%$  in the group of patients with ANCA-positive IgA nephropathy (Table 3; Fig 3). About one-fifth of crescents were fibrous. Only borderline or mild mesangial hypercellularity was observed. Endocapillary hypercellularity was present in 4 patients. Segmental tuft necrosis was observed in 4 patients. Synechiae also were common. Mild to moderate or even severe tubular atrophy was accompanied by different degrees of interstitial fibrosis. In the lumen of many tubules, erythrocytes or protein casts were seen. In immunohistochemistry, all 8 patients showed dominant positive staining for IgA in the mesangium (range, + to ++ on a scale of 0 to +++; Fig 4). IgG was positive in 3 patients; C3, in 6 patients; C1q, in 5 patients; and IgM and fibrin in all 8 patients (Table 3). Positive staining for IgM and fibrin in crescents was observed in 4 patients. Fibrin was found within synechiae and/or tuft necrosis in all cases, and in half of these patients, in combination with IgM. No capillary staining was observed. In electron microscopy, in addition to electron-dense deposits in the mesangial matrix, focal or diffuse loss of podocyte processes was observed. No subendothelial or subepithelial deposits were found.

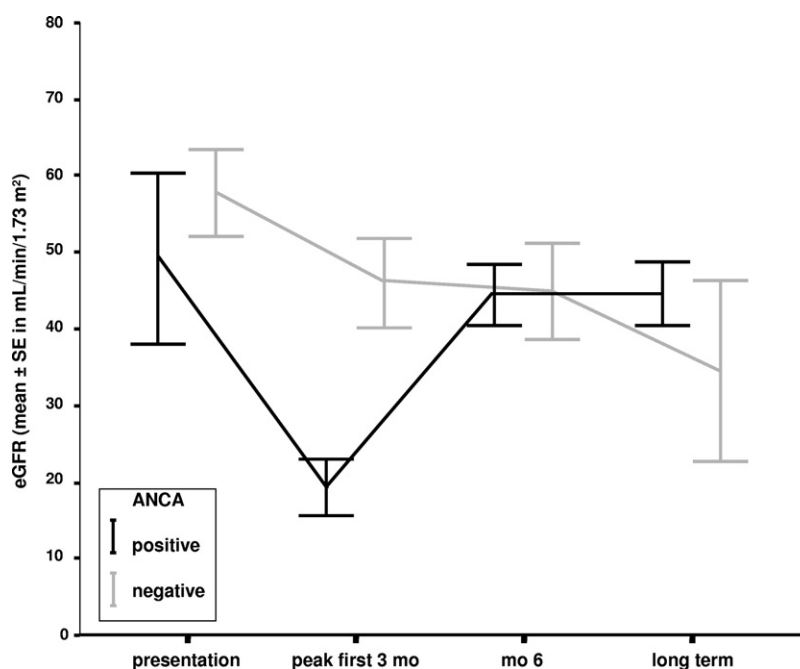


**Figure 1.** Course of kidney function based on serum creatinine level in patients with crescentic immunoglobulin A nephropathy according to the presence ( $n = 8$ ) or absence ( $n = 26$ ) of antineutrophil cytoplasmic autoantibody (ANCA). Abbreviation: SE, standard error.

#### Comparison With Patients With ANCA-Negative Crescentic IgA Nephropathy

Compared with patients with ANCA-negative crescentic IgA nephropathy, ANCA-positive patients had a more severe histologic picture with a higher percentage of crescentic glomeruli ( $54.3\% \pm 18\%$  vs  $34.5\% \pm 26\%$ ). Crescents in  $> 50\%$  of glomeruli were observed in  $75.0\%$  of ANCA-positive compared with  $30.8\%$  of ANCA-nega-

tive patients. Furthermore, mesangial hypercellularity was markedly increased in ANCA-negative (+,  $40\%$ ; ++,  $52\%$ ; and +++,  $2\%$ ) compared with ANCA-positive patients (+,  $100\%$ ). The intensity of mesangial IgA deposits was similar between ANCA-negative (+,  $52\%$ ; ++,  $40\%$ ; and +++,  $8\%$ ) and ANCA-positive patients (+,  $50\%$ ; ++,  $50\%$ ). In contrast to ANCA-positive patients, IgA deposits along the capillary wall



**Figure 2.** Course of kidney function based on estimated glomerular filtration rate (eGFR) calculated using the Modification of Diet in Renal Disease Study equation in patients with crescentic immunoglobulin A nephropathy according to the presence ( $n = 8$ ) or absence ( $n = 26$ ) of antineutrophil cytoplasmic autoantibody (ANCA). Abbreviation: SE, standard error.

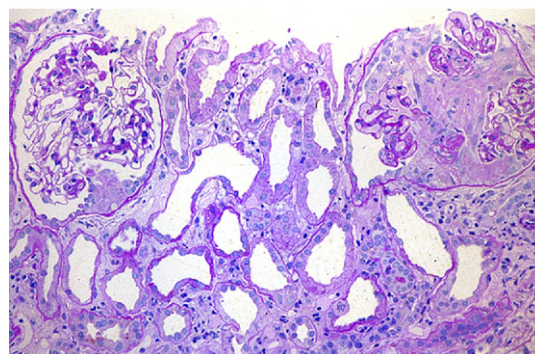
**Table 2.** Clinical Characteristics of Patients With Crescentic Immunoglobulin A Nephropathy According to the Presence or Absence of ANCAs

|                                                    | ANCA Positive | ANCA Negative |
|----------------------------------------------------|---------------|---------------|
| No. of patients                                    | 8             | 26            |
| Age (y)                                            | 56.0 ± 6.3    | 34.5 ± 12.7   |
| Initial SCr (mg/dL)                                | 2.7 ± 2.6     | 1.9 ± 1.4     |
| Initial eGFR (mL/min/1.73 m <sup>2</sup> )         | 49.3 ± 31.7   | 57.8 ± 28.9   |
| Peak SCr (mg/dL)                                   | 4.2 ± 2.2     | 2.5 ± 1.9     |
| Lowest eGFR (mL/min/1.73 m <sup>2</sup> )          | 19.3 ± 10.2   | 45.9 ± 30.1   |
| Proteinuria (g/24 h)                               | 3.9 ± 2.4     | 3.2 ± 2.8     |
| Hemoglobin (g/dL)                                  | 10.5 ± 2.0    | 12.7 ± 1.5    |
| Mean arterial blood pressure (mm Hg)               | 105.5 ± 8.1   | 104.2 ± 16.7  |
| Macrohematuria (%)                                 | 37.5          | 23.1          |
| Systemic symptoms (%)                              | 87.5          | 38.5          |
| Rapidly deteriorating kidney function (%)          | 100           | 36            |
| Stable or gradually decreasing kidney function (%) | —             | 64            |

Note: Conversion factors for units: SCr in mg/dL to  $\mu\text{mol/L}$ ,  $\times 88.4$ ; eGFR in mL/min/1.73 m<sup>2</sup> to mL/s/1.73 m<sup>2</sup>,  $\times 0.01667$ .

Abbreviations and definitions: ANCA, antineutrophil cytoplasmic autoantibody; eGFR, estimated glomerular filtration rate calculated using the 4-variable Modification of Diet in Renal Disease Study equation; SCr, serum creatinine.

(supepithelial) were found in 11 ANCA-negative patients (42%). Endocapillary hypercellularity was found in 12 ANCA-negative patients, and segmental tuft necrosis, in 3. ANCA-positive

**Figure 3.** Photomicrograph of the kidney biopsy specimen from patient 4. Fibrocellular crescents are seen in both glomeruli, together with mild mesangial hypercellularity. Severe mesangial matrix expansion and a synechia in Bowman capsule also are evident (Courtesy of U. Helmchen, MD; Department of Pathology, Hamburg University).

patients had a higher degree of tubular atrophy ( $43\% \pm 20\%$ ) compared with ANCA-negative patients ( $28\% \pm 21\%$ ).

### Response to Treatment

#### ANCA-Positive Patients With IgA Nephropathy

All patients with ANCA-positive IgA nephropathy were treated with a combination of cyclophosphamide and corticosteroids (intravenous therapy,  $n = 4$ ; oral therapy,  $n = 4$ ) for 6 months (intravenous therapy: cyclophosphamide, 500 mg/m<sup>2</sup> body surface every 4 weeks; oral therapy: 2 mg/kg). As maintenance therapy, oral cyclophosphamide ( $n = 2$ ), azathioprine ( $n = 4$ ), or mycophenolate mofetil ( $n = 2$ ) were used. No patient required dialysis during the acute phase,

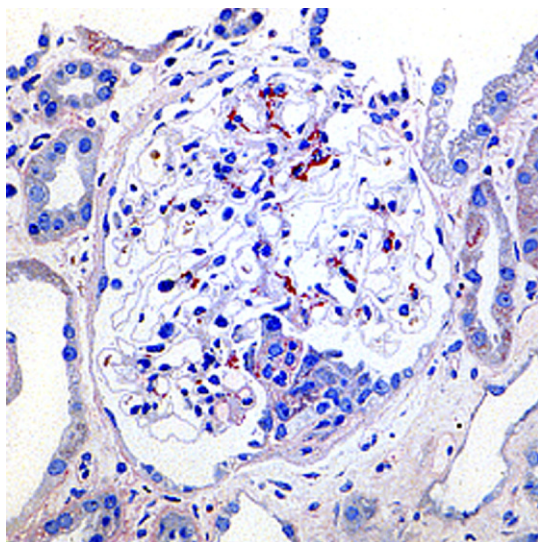
**Table 3.** Histologic Characteristics of ANCA-Positive Patients With IgA Nephropathy

| Patient No. | ANCA | Global Sclerosis (%) | Cellular Crescents (%) | Fibrous Crescents (%) | Mesangial Hyperplasia | Tubular Atrophy (%) | Interstitial Infiltration | IgA | IgG | IgM | C3 | C1q | Fibrin |
|-------------|------|----------------------|------------------------|-----------------------|-----------------------|---------------------|---------------------------|-----|-----|-----|----|-----|--------|
| 1           | MPO  | 0                    | 29                     | 29                    | +                     | 30-40               | +                         | +   | 0   | +   | +  | +   | +      |
| 2           | MPO  | 33                   | 25                     | 0                     | +                     | 70-80               | 0                         | +   | 0   | +   | 0  | +   | +      |
| 3           | MPO  | 0                    | 33                     | 33                    | +                     | 60                  | 0                         | +   | 0   | +   | +  | 0   | +      |
| 4           | MPO  | 0                    | 29                     | 43                    | +                     | 30-40               | 0                         | ++  | 0   | +   | +  | +   | +      |
| 5           | MPO  | 19                   | 30                     | 30                    | +                     | 30                  | 0                         | ++  | +   | +   | +  | +   | +      |
| 6           | PR3  | 14                   | 0                      | 28                    | +                     | 30-40               | 0                         | ++  | +   | +   | +  | 0   | +      |
| 7           | PR3  | 0                    | 60                     | 0                     | +                     | 60                  | 0                         | +   | 0   | +   | 0  | 0   | +      |
| 8           | PR3  | 0                    | 67                     | 0                     | +                     | 10-20               | 0                         | ++  | +   | +   | +  | +   | +      |

Note: Histologic characteristics are on a scale of 0-3: 0 for absence, + for mild, ++ for moderate, and +++ for severe/strong.

Abbreviations: ANCA, antineutrophil cytoplasmic antibody; eGFR, estimated glomerular filtration rate; IgA, immunoglobulin A; MPO, myeloperoxidase; PR3, proteinase 3; SCr, serum creatinine.





**Figure 4.** Photomicrograph of immunohistochemistry of patient 4. Immunoglobulin A is deposited in the mesangium of a glomerulus that also shows a small fibrocellular crescent (Courtesy of U. Helmchen, MD; Department of Pathology, Hamburg University).

and test results for ANCAs became negative in all cases. As shown in Figs 1 and 2, kidney function improved in all patients: serum creatinine level decreased from a peak of  $4.2 \pm 2.2$  to  $1.7 \pm 0.3$  mg/dL 6 months after the initiation of treatment (MPO-ANCA,  $4.8 \pm 2.5$  to  $1.7 \pm 0.37$  mg/dL; PR3-ANCA,  $3.1 \pm 1.0$  to  $1.6 \pm 0.2$  mg/dL), and eGFR improved from  $19.3 \pm 10.2$  to  $44.4 \pm 11.1$  mL/min/1.73 m<sup>2</sup> (MPO-ANCA,  $16.2 \pm 10.7$  to  $40.4 \pm 11.6$  mL/min/1.73 m<sup>2</sup>; PR3-ANCA,  $24.4 \pm 8.6$  to  $50.9 \pm 7.6$  mL/min/1.73 m<sup>2</sup>). Kidney function was stable in the long term (serum creatinine,  $1.7 \pm 0.7$  mg/dL [MPO-ANCA,  $1.5 \pm 0.3$  mg/dL; PR3-ANCA,  $2.2 \pm 1.1$  mg/dL], and eGFR,  $44.6 \pm 11.7$  mL/min/1.73 m<sup>2</sup> [MPO-ANCA,  $46.8 \pm 7.8$  mL/min/1.73 m<sup>2</sup>; PR3-ANCA,  $40.9 \pm 17.9$  mL/min/1.73 m<sup>2</sup>]). Mean proteinuria was decreased from the initial protein excretion of  $3.9 \pm 2.4$  to  $2.7 \pm 1.7$  g/24 h in 6 months to  $1.8 \pm 2.5$  g/24 h in the long term. Hemoglobin levels increased from a minimum of  $10.5 \pm 2.0$  g/dL before initiation of treatment to  $12.8 \pm 1.5$  g/dL after 6 months and  $14.1 \pm 1.1$  g/dL in the long term.

In 4 patients, subsequent biopsy specimens after treatment were available. In these biopsy specimens, no tuft necrosis, synechiae, or crescents were observed. However, the number of

sclerosed glomeruli and degree of tubular atrophy and interstitial fibrosis increased. In immunohistochemistry, IgA was still present in the mesangium.

#### **Comparison With Patients With ANCA-Negative Crescentic IgA Nephropathy**

Twelve patients with ANCA-negative crescentic IgA nephropathy received immunosuppressive treatment (cyclophosphamide and corticosteroids,  $n = 5$ ; steroids only,  $n = 5$ ; mycophenolate mofetil,  $n = 2$ ). The rest of the patients were treated with angiotensin-converting enzyme inhibitors, and a few were treated with fish oil. In contrast to ANCA-positive patients, no significant improvement in kidney function was observed after 6 months of treatment. In long-term follow-up, ANCA-negative patients with crescentic IgA nephropathy showed further deterioration in kidney function (Figs 1 and 2, grey lines).

### **DISCUSSION**

In the present report, we describe the clinical and histologic course of 8 patients with crescentic IgA nephropathy and coexisting positive test results for ANCAs. Their data were compared with those for 26 patients with crescentic IgA nephropathy, but no ANCAs. ANCA-positive patients presented with a more severe clinical picture, with rapidly deteriorating kidney function, general symptoms, and more serious histologic lesions, and showed a different response to treatment, giving overall the impression of a novel clinical entity.

ANCAs are autoantibodies against components of the cytoplasm of neutrophils. The most common antigen specificities are for PR3 and MPO. The sensitivity and specificity of the array used to detect ANCAs varies. Indirect immunofluorescence has higher sensitivity, but lower specificity, because apart from antibodies against PR3 or MPO, it also can detect antibodies against other antigens, including lactoferrin, elastase, and cathepsin G. ELISA provides a greater degree of specificity, but is less sensitive. Combined analysis using indirect immunofluorescence and ELISA leads to 99% specificity.<sup>16</sup>

ANCAs are associated with a spectrum of diseases, including Wegener granulomatosis, microscopic polyangiitis, Churg-Strauss syndrome, and isolated pauci-immune crescentic glomerulo-

**Table 4.** Overview of the Association of Immunoglobulin A Nephropathy and ANCA Autoantibodies as Reported in the Literature

| Study                          | No. of Patients | Sex      | Age (y) | ANCA                 | Pulmonary Involvement | Follow-up (mo) | Dialysis on Presentation | Treatment                             | Long-term Dialysis | Remarks                                       |
|--------------------------------|-----------------|----------|---------|----------------------|-----------------------|----------------|--------------------------|---------------------------------------|--------------------|-----------------------------------------------|
| O'Donoghue et al <sup>16</sup> | 2               | NR       | NR      | 2 MPO                | 0                     | NR             | 0                        | —                                     | 0                  | Slow progressive kidney failure, no crescents |
| Allmaras et al <sup>7</sup>    | 3               | 2 M, 1 F | 62 ± 7  | 3 MPO                | 2                     | 10 ± 1         | 2                        | Corticoids IV, CP IV (plasmapheresis) | 0                  | —                                             |
| Martin et al <sup>8</sup>      | 1               | M        | 68      | MPO                  | No                    | 15             | No                       | Corticoids IV                         | Yes                | Relapse in kidney transplant                  |
| McGregor et al <sup>9</sup>    | 1               | M        | 14      | MPO                  | Yes                   | 12             | No                       | Corticoids IV, CP IV                  | No                 | —                                             |
| Ramirez et al <sup>10</sup>    | 1               | F        | 10      | MPO                  | No                    | 10             | Yes                      | Corticoids IV, CP IV                  | Yes                | A single pulse                                |
| Haas et al <sup>11</sup>       | 6               | 3 M, 3 F | 48 ± 21 | 1 MPO, 4 PR3, 1 both | 3                     | 16 ± 11        | 3                        | Corticoids, CP                        | 2                  | —                                             |
| Tagboto et al <sup>12</sup>    | 1               | F        | 40      | MPO                  | No                    | 2              | No                       | Corticoids orally, CP orally          | No                 | Coexisting Hermansky-Pudlak syndrome          |
| Lui et al <sup>13</sup>        | 1               | F        | 72      | MPO                  | No                    | 7              | No                       | Corticoids orally, CP orally          | No                 | Coexisting diabetic nephropathy               |
| Jimbo et al <sup>14</sup>      | 1               | F        | 14      | MPO                  | No                    | 30             | No                       | Corticoids IV                         | No                 | —                                             |
| Ogawa et al <sup>15</sup>      | 1               | F        | 20      | MPO                  | No                    | 56             | No                       | Corticoids IV, tonsillectomy          | No                 | —                                             |
| Present study                  | 8               | 6 M, 2 F | 56 ± 6  | 5 MPO, 3 PR3         | 5                     | 29 ± 20        | 0                        | Corticoids, CP                        | 0                  | —                                             |

Abbreviations: ANCA, antineutrophil cytoplasmic antibody; CP, cyclophosphamide; IV, intravenously; MPO, myeloperoxidase; NR, not reported; PR3, proteinase 3.

nephritis. There is an increasing body of evidence that ANCAs are not merely markers, but also are involved in the pathogenesis of these diseases.<sup>17</sup> In many cases, ANCA titers correlate with disease activity.<sup>18</sup>

In the present study, we describe a cohort of 8 patients with IgA nephropathy positive for ANCAs (5 positive for ANCA against MPO and 3 positive for ANCA against PR3). All ANCA-positive patients with IgA nephropathy presented with rapidly decreasing kidney function and active urinary sediment. Mean hemoglobin levels were lower than normal, and all except 1 had general symptoms, such as weakness, arthralgias, and fever. ANCA-positive patients with IgA nephropathy had a higher percentage of crescentic glomeruli and higher degree of tubular atrophy compared with ANCA-negative patients with crescentic IgA nephropathy. The intensity of mesangial IgA deposits was similar between ANCA-positive and ANCA-negative patients, but mesangial hypercellularity was markedly increased in ANCA-negative compared with ANCA-positive patients. In contrast to ANCA-negative patients, endocapillary hypercellularity was a constant feature in ANCA-positive patients. However, no IgA deposits along the capillary wall (subendothelial) were observed in these patients (in contrast to ANCA-negative patients).

There are few previous reports of positive ANCAs in patients with IgA nephropathy, mainly case reports<sup>6,8-10,12-15</sup> and 2 case series<sup>7,11</sup> (Table 4). With the exception of the cases reported by O'Donoghue et al,<sup>6</sup> all other investigators observed the clinical presentation of rapidly deteriorating kidney function accompanied with nephritic sediment and the histologic picture of crescentic glomerulonephritis. Of interest, 5 of the 9 patients reported by the case series and 1 of 9 case reports had extrarenal manifestations, with pulmonary involvement (hemoptysis/infiltrates) as a hallmark. In concordance with our results, all patients reported in the larger series by Haas et al<sup>11</sup> had > 50% crescents, but only mild mesangial hypercellularity.

Without doubt, it is a great challenge to interpret the coexistence of mesangial hypercellularity and matrix expansion, intra- and extracapillary glomerulonephritis, mesangial IgA deposits, and ANCA autoantibodies. The possibility that these patients developed coincidental ANCA-

associated crescentic glomerulonephritis on top of preexisting IgA nephropathy could be one reasonable explanation. IgA deposition is common in the general population, and only a few patients develop clinical overt IgA nephropathy. Mesangial IgA deposition was detected in 4.8% of autopsy cases without a history of kidney disease or with only minor urine abnormalities.<sup>19</sup> Therefore, the possibility of preexisting not previously diagnosed IgA nephropathy that has been superimposed by ANCA-positive crescentic glomerulonephritis cannot be excluded. Extrarenal manifestations, including respiratory tract symptoms, are not typical features of IgA nephropathy, and their presence could be explained better by a coexisting ANCA-associated vasculitis. However, respiratory involvement, such as pulmonary hemorrhage, is a rare manifestation of IgA nephropathy, but is described in even ANCA-negative patients.<sup>20,21</sup> The exceptionally good prognosis with aggressive treatment (discussed next) compared with classic ANCA-associated vasculitis also is an argument against this theory. Furthermore, this possibility seems unlikely based on the frequency of this "coincidence" in our collective.

One also could argue that these simply are patients with crescentic IgA nephropathy (a well-documented variant of IgA nephropathy) and that the presence of ANCA autoantibodies is merely a coincidence, a laboratory finding without pathogenetic and clinical impact. This theory could be supported by the study of O'Donoghue et al,<sup>6</sup> who screened 100 patients with IgA nephropathy and found 2 patients who were positive for ANCAs, but had slowly decreasing kidney function and no crescents. However, our study and the study of Haas et al<sup>11</sup> compared 8 and 6 ANCA-positive patients with 26 and 8 ANCA-negative patients with crescentic IgA nephropathy, respectively. Both observed a distinguishable clinical and histologic picture, suggesting that the presence of ANCAs has an effect. Martin et al<sup>8</sup> reported recurrence of crescentic IgA nephropathy associated with ANCA autoantibodies in a kidney transplant. Of interest, ANCA titers correlated with the clinical picture and increased during the relapse in the kidney transplant, suggesting a pathogenetic role.<sup>8</sup>

More attractive therefore appears the possibility that the coexistence of ANCAs and crescentic



IgA nephropathy represents a novel variant of crescentic IgA nephropathy, possibly an overlap syndrome of IgA nephropathy with ANCA-positive vasculitis. ANCA also can be present in about one-third of patients with anti-glomerular basement membrane disease. Similar to our findings in patients with IgA nephropathy, this double positivity influences patient outcome, possibly defining a genuine disease.<sup>22</sup>

Notably all ANCA-positive patients with IgA nephropathy in our cohort and most patients reported by other investigators responded very well to immunosuppressive treatment, similar to regimens used in patients with classic ANCA-positive vasculitis with renal involvement. All 3 patients reported by Allmaras et al benefited from cyclophosphamide and corticosteroids, and 2 patients who were initially on hemodialysis therapy regained kidney function and dialysis therapy could be terminated.<sup>7</sup> Haas et al also treated all their patients with cyclophosphamide and corticosteroids. Two patients were already on dialysis therapy at the time of kidney biopsy and did not respond to treatment, 3 patients improved, and 1 patient was lost to follow up.<sup>11</sup> All patients reported as case reports received immunosuppressive treatment, and in most cases, kidney function could be preserved (Table 4). Our ANCA-positive patients with IgA nephropathy also were treated with a combination of cyclophosphamide and corticosteroids. Kidney function improved in all patients, and mean proteinuria was decreased. In 4 patients, subsequent biopsies were available in which no tuft necrosis, synechiae, or crescents were found, but glomerular sclerosis, tubular atrophy, and interstitial fibrosis had increased. In contrast, no significant improvement in kidney function was achieved in ANCA-negative patients, and further deterioration of kidney function was observed in long-term follow-up. However, one could argue that the progressive course of ANCA-negative crescentic patients could be attributed to the lack of aggressive treatment: Because most presented with the clinical picture of chronic glomerulonephritis, the attending physicians often refrained from immunosuppression. Administration of corticosteroids and intravenous cyclophosphamide for 6 months in (ANCA-negative) patients with > 10% crescents decreased peak serum creatinine level,

proteinuria, and the rate of progression to kidney failure compared with historical controls.<sup>5</sup>

In the present study, we describe a subset of patients with crescentic IgA nephropathy and ANCA antibodies characterized by a diverse more exaggerated clinical and histologic picture. Of particular interest is that although ANCA-positive patients presented with rapidly deteriorating kidney function and more serious histologic lesions, their response to aggressive immunosuppressive treatment was excellent. ANCA antibodies therefore should be determined in all patients with IgA nephropathy because their presence appears to influence patient outcome and response to treatment, and this association might confer distinct prognostic and therapeutic implications.

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